

KUBERNETES

1.What is Kubernetes ?

Kubernetes is an open-source container orchestration platform used to manage containerized applications. It automates deployment, scaling, networking, and management of containers and helps maintain application availability and health by monitoring and replacing unhealthy Pods.

2.Components of Kubernetes :

Kubernetes mainly contains two components and they are control plane and worker node.

Control Plane Components : (Master Node)

- API server
- etcd
- Kube Scheduler
- Control Manager

Worker Node Components :

- Kubelet
- Container Runtime
- Kube Proxy

Control Plane :

1.API Server :

The API Server is the entry point to the Kubernetes cluster. It authenticates and authorizes requests, validates Kubernetes objects, and communicates with other control-plane components.

2.etcd :

etcd is a distributed key-value database that stores the Kubernetes cluster's persistent state and configuration.

3.Kube Scheduler :

The kube-scheduler selects a suitable worker node for newly created Pods based on resource requirements, node availability, affinity/anti-affinity, taints and tolerations, and other scheduling rules.

4. Control Manager :

The Controller Manager runs controllers that continuously compare the desired state with the current state and take actions to bring the cluster back to the desired state.

Worker Node :

1. Kubelet :

The kubelet is an agent running on every worker node. It communicates with the API Server and ensures that the Pods assigned to that node are running according to their specifications.

2. Container Runtime :

The container runtime is responsible for pulling container images and creating, starting, stopping, and managing containers.

3. Kube Proxy :

kube-proxy maintains network rules on nodes that help implement Kubernetes Services and direct network traffic to the appropriate backend Pods.

3. Explain how pods, services, deployments, and namespaces work together.

Namespace organizes the resources, Deployment manages the Pods, Pods run the application containers, and Service provides stable network access to those Pods.

Namespace :

Namespaces are used to logically organize and separate Kubernetes resources within a cluster.

Deployments :

Deployments manage the desired number of Pods and ensure that the required number of replicas are running. They also handle rolling updates and help maintain the desired state.

Pods :

Pods are the smallest deployable units in Kubernetes and contain one or more containers where the application actually runs.

Services :

Services provide stable network access to Pods. Since Pod IP addresses can change when Pods are recreated, a Service provides a stable IP/DNS name and routes traffic to the appropriate Pods using labels and selectors.

3.How different components in Kubernetes interact with each other ?

1. User sends a request using kubectl commands.
2. API Server receives the request and authenticates, authorizes, and validates it.
3. API Server stores the desired state in etcd.
4. Controller Manager watches the cluster state and checks whether the actual state matches the desired state.
5. Scheduler identifies newly created Pods that don't have a node assigned.
6. Scheduler selects a suitable Worker Node based on resources and scheduling rules.
7. API Server records the Pod-to-node assignment.
8. Kubelet on the selected Worker Node notices the assigned Pod.
9. Kubelet communicates with the Container Runtime to create and start the container.
10. Container Runtime pulls the image if it is not already available and runs the container.
11. Kubelet continuously monitors the Pod and reports its status to the API Server.
12. Controller Manager maintains the desired state. If a Pod fails, it works with the other components to create a replacement.
13. kube-proxy manages Service networking and helps route traffic from a Kubernetes Service to the appropriate Pods.

4. Architecture of Kubernetes :

